Regression (variance of ℓ_2 loss)

Var of ℓ_2 loss (relative to $t=0$)

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2

$3\times$
$3.5\times$
$2.5\times$

0
2
4
6
8
10

Regression (RMSE of estimated ℓ_2 loss)

RMSE of estimated ℓ_2 loss (relative to $t=0$)

1.0
0.8
0.6
0.4

0
2
4
6
8
10

Classification (variance of log-loss)

Var of log-loss (relative to $t=0$)

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3

$3\times$
$2.3\times$
$2.7\times$

0
2
4
6
8
10

Decision epoch t

Classification (RMSE of estimated log-loss)

RMSE of estimated log-loss (relative to $t=0$)

1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4

0
2
4
6
8
10

Decision epoch t

AUTODIFF 1-lookahead
REINFORCE (with baseline) 1-lookahead
REINFORCE (without baseline) 1-lookahead
Random sampling
Uncertainty sampling - static
Uncertainty sampling - sequential